Harvey **Mannering**

PhD Researcher · Generative Models · Ethical AI

harveymannering@gmail.com
0754 366 4176
harveymannering.github.io
Google Scholar
harvey-mannering
harveymannering

SUMMARY

Final year PhD student at the University of Southampton, with publications at leading machine learning conferences such as **NeurIPS**, **WACV**, and **ICML**. My research interests are around ethical AI, generative models (in particular diffusion models) and affective computing. Prior to my PhD I gained four years experience as a software developer (including UCL's Advanced Research Computing Center). Currently seeking research internships.

KEY PUBLICATIONS

WACV 2027 **Mannering, H.**, Huang, Z., and Prügel-Bennett, A. *Attractive Faces Skew Text-to-Image Reward Models and Leaderboards.*

NeurIPS 2026 **Mannering, H.**, Huang, Z., and Prügel-Bennett, A. *Barycentric Guidance: Turning Foundation Image Editors into Continuous Affective Controllers.*

ICML 2025 Smith, D., **Mannering, H.** and Marcu, A. *Functional Alignment Can Mislead: Examining Model Stitching.* **Spotlight**

ICLP 2023 **Mannering, H.** *Analysing Gender Bias in Text-to-Image Models using Object Detection.* Safe and Trustworthy AI Workshop.

EDUCATION

University of Southampton | *PhD Computer Science* 2024 – Present

- **Areas of Interest** Safe and Ethical AI, Generative AI, Diffusion Models, Affective Computing, Human-centered AI
- **Technical Skills** PyTorch · Diffusers · LoRA Fine-tuning · Text-to-Image Models · Image Editing Models · Multi-GPU Training · Reward Models · Python · Git

UCL | *MSc Computer Graphics, Vision and Imaging · Distinction* 2021 – 2022

- **Project** Glass detection in RGB video via segmentation and active learning methods.
- **Specialist Areas** Deep Learning, Computer Vision, Machine Learning for Medical Imaging, Image Processing, Computer Graphics, Numerical Optimisation, MatLab, Python, PyTorch

University of Southampton | *BSc Computer Science · 2.1* 2014 – 2017

- **Project** Converted audio signals into sheet music using signal processing and machine learning.
- **Specialist Areas** Computer Vision, Machine Learning, Signal Processing, Databases, Computer Graphics, Embedded Systems, Java, C, C++, MatLab

EXPERIENCE

UCL | *Research Software Engineer* London, England · Oct 2022 – Sep 2023

- Collaborated with researchers on healthcare tools, including a real-time hospital database for machine learning research. Learnt best practices for research software development.

PE.Fiberoptics | *Software Engineer* Wokingham, England · Jan 2020 – Sep 2021

- Worked as the R&D department's main software engineer. Used C# and WPF to implement signal processing and image processing algorithms. Implemented and managed version control for the company.

Gamma Telecoms | *Software Developer/Analyst* Newbury, England · Aug 2017 – Mar 2019

- Gained direct experience with Java, MS SQL, Git, HTML, CSS, JavaScript and Jenkins. Full stack developer collaborating with seven other developers within an agile framework.